

Module 2: Monte Carlo Methods

MA 563 – Statistical Computation and Simulation

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Unit Summary

A Problem That Shows Up Everywhere

- Across science and engineering, we constantly need to evaluate quantities of the form

$$\theta = \int_{\mathcal{X}} h(x) f(x) dx = E_f[h(X)],$$

$$\Rightarrow X \sim f$$
$$\checkmark E(x) = \int x f_x(x) dx$$

where f is a probability density (or an un-normalized weight) and h is some function of interest.

$$E(h(x)) = \int h(x) f(x) dx$$

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The central question of Module 2

Can we estimate $\theta = E_f[h(X)]$ *without* ever solving the integral – using nothing but random samples?

Example: A Simple Integral With No Analytic Solution

- Consider the innocent-looking integral

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Example: A Simple Integral With No Analytic Solution

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$$\theta = \int_0^1 \left(\frac{\sin x}{x} \right) dx.$$

Handwritten notes in red:

- $h(u) \Rightarrow = \int_0^1 h(u) 1 du$
- $E_f(h(u)) \Rightarrow f \sim U[0,1]$
- $\Rightarrow U_1, U_2, \dots, U_n \sim U[0,1]$
- $= \frac{1}{n} \sum_{i=1}^n h(U_i)$
- $= \frac{1}{n} \sum_{i=1}^n \frac{\sin U_i}{U_i}$

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But Monte Carlo does not care

Write $\theta = E[h(U)]$ with $U \sim \text{Unif}(0, 1)$ and $h(u) = \sin(u)/u$ (defining $h(0) = 1$ by continuity). Draw $U_1, \dots, U_n$ and average $h(U_i)$ – exactly as easy as any other integral, whether or not a closed form exists.

Example: Integrals Famous Enough to Get Their Own Name

- Some integrals are so common, yet so resistant to elementary evaluation, that mathematicians simply **defined a new function** to be the answer:

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx, \quad B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

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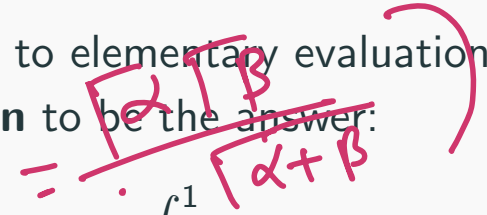
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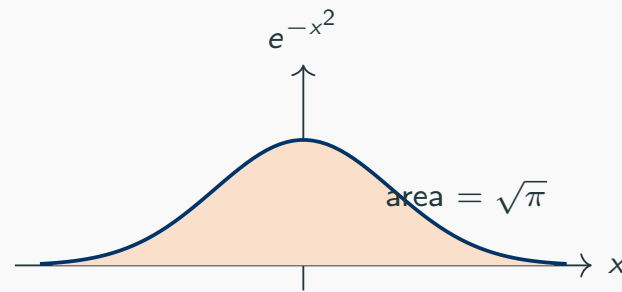
The honest picture

Even values you take for granted, like $\Gamma(1/2) = \sqrt{\pi}$, come from evaluating the integral cleverly at that *one* special point – for a generic α (e.g. $\Gamma(2.37)$), there is no shortcut at all.

Example: A Famous Integral – Try It Analytically First

- Before turning to genuinely hopeless integrals, try this classic one **by hand**:

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$



✓
 $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \sim \text{pdf of } N(0,1)$

$$I_x = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx$$

$$I_y = \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy$$

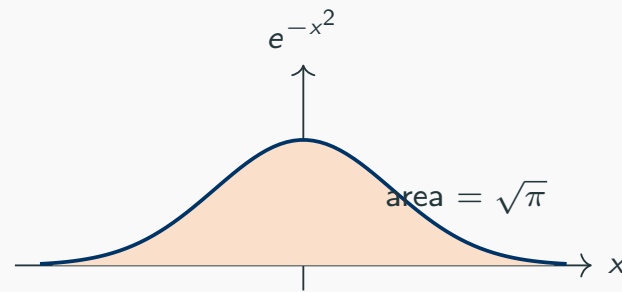
$$\Rightarrow I_x = I_y \quad \Rightarrow I_x \cdot I_y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dx dy$$

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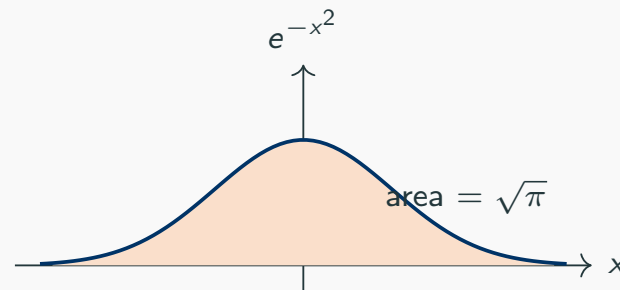
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- Yet I has an exact closed form! The classical trick: square it and switch to polar coordinates,

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = \pi, \quad \text{giving } I = \sqrt{\pi}.$$



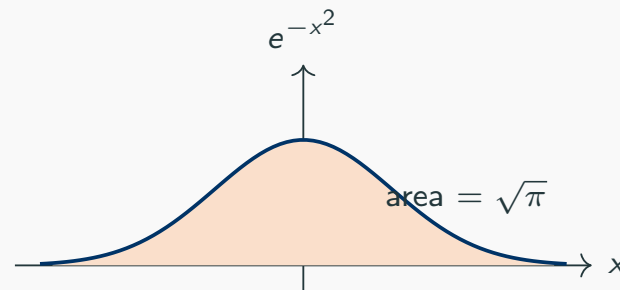
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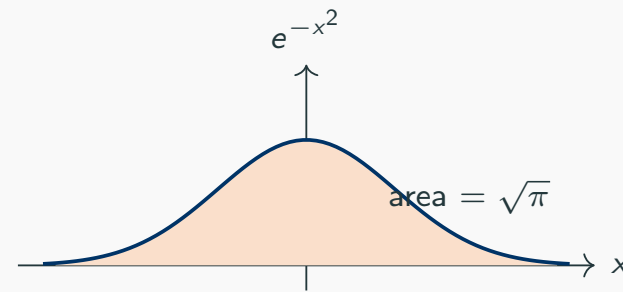
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Classroom challenge

Work out $I = \sqrt{\pi}$ analytically using the squaring-and-polar-coordinates trick above – it is the same polar-coordinates idea behind the Box-Muller transform from Module 1! Then compare it with a Monte Carlo estimate of I once we build the machinery in this module.

Example: Pricing a Path-Dependent Option (Finance)

- An **Asian option** pays off based on the *average* stock price over its life, not just the final price:

$$\text{Payoff} = \max(\bar{S} - K, 0), \quad \bar{S} = \frac{1}{T} \int_0^T S_t dt.$$

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Standard practice on every trading floor

Simulate thousands of possible price paths, compute the payoff on each, and average – a direct application of the Monte Carlo principle we are about to formalize.

Example: Bayesian Evidence in Genomics and AI

- In Bayesian inference (genomics: estimating population parameters from SNP data; AI: Bayesian neural networks), the posterior mean of a quantity $h(\theta)$ is

$$E[h(\theta) \mid \text{data}] = \frac{\int h(\theta) L(\text{data} \mid \theta) \pi(\theta) d\theta}{\int L(\text{data} \mid \theta) \pi(\theta) d\theta}.$$

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- Both numerator and denominator (the **marginal likelihood** or “evidence”) are integrals over $\theta \in \mathbb{R}^d$, often with d in the hundreds or thousands (e.g. millions of genetic markers, or network weights).
- Classical quadrature is hopeless in such dimensions – this single fact motivates almost all of modern computational statistics.

The Curse of Dimensionality

- A classical quadrature rule (e.g. Simpson's rule) evaluates h on a regular grid.

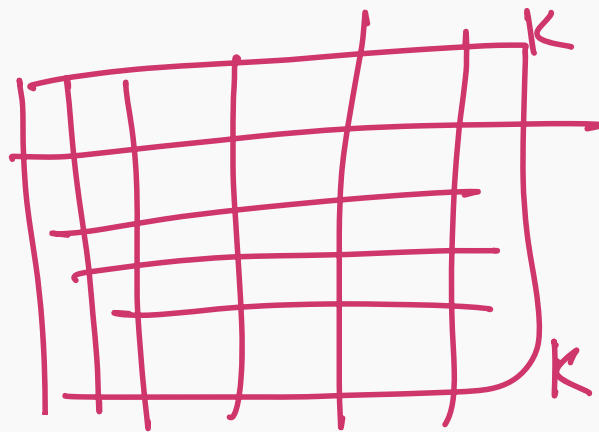
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- To keep the same accuracy in d dimensions, we need roughly k^d grid points, where k is the number of points needed per dimension.

$$k^2$$

$$k^d$$

$$k =$$



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Preview of the punchline

We will show that a Monte Carlo estimator's error shrinks at rate $O(n^{-1/2})$ **regardless of dimension d** . This dimension-free property is precisely why simulation, not quadrature, is the workhorse of high-dimensional computation.

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